

Introduction to Analysis II HW8

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May 3, 2026

Problem 1. Let $p > 2$ and $c > 0$. Using the Borel–Cantelli lemma, show that the set

$$\left\{x \in [0, 1] : \left|x - \frac{a}{q}\right| \leq \frac{c}{q^p} \text{ for infinitely many positive integers } a, q\right\}$$

has measure zero. (Hint: one only has to consider those integers a in the range $0 \leq a \leq q$ (why?). Use Corollary 11.6.5 to show that the sum $\sum_{q=1}^{\infty} \frac{c(q+1)}{q^p}$ is finite.)

Proof. Let

$$E_{a,q} := \left\{x \in [0, 1] : \left|x - \frac{a}{q}\right| \leq \frac{c}{q^p}\right\},$$

then for each positive integer a, q , we know $E_{a,q}$ is a one-dimensional closed box of volume $\frac{2c}{q^p}$, so $m(E_{a,q}) = \frac{2c}{q^p}$. Also, each $E_{a,q}$ is measurable since it is a closed set. Now define $\Omega_q = \bigcup_{a=1}^q E_{a,q}$ for each positive integer q , then we have

$$m(\Omega_q) \leq \sum_{a=1}^q m(E_{a,q}) = \sum_{a=1}^q \frac{2c}{q^p} = \frac{2cq}{q^p} \quad \forall q \in \mathbb{N}.$$

Hence,

$$\sum_{q=1}^{\infty} m(\Omega_q) \leq \sum_{q=1}^{\infty} \frac{2cq}{q^p} = 2c \sum_{q=1}^{\infty} \frac{1}{q^{p-1}} < \infty$$

since $p - 1 > 1$ and by p-series test this is true. Now by Borel-Cantelli lemma, we have

$$m(\{x \in \mathbb{R} : x \in \Omega_q \text{ for infinitely many } q\}) = 0.$$

Note that

$$\begin{aligned} E &:= \{x \in \mathbb{R} : x \in \Omega_q \text{ for infinitely many } q\} \\ &= \left\{x \in \mathbb{R} : x \in \bigcup_{a=1}^q E_{a,q} \text{ for infinitely many } q\right\} \\ &= \{x \in \mathbb{R} : x \in E_{a,q} \text{ for infinitely many positive integer pairs } (a, q) \text{ s.t. } a \leq q\} \\ &= \left\{x \in [0, 1] : \left|x - \frac{a}{q}\right| \leq \frac{c}{q^p} \text{ for infinitely many positive integer pairs } (a, q) \text{ s.t. } a \leq q\right\}, \end{aligned}$$

and let

$$F := \left\{x \in [0, 1] : \left|x - \frac{a}{q}\right| \leq \frac{c}{q^p} \text{ for infinitely many positive integers } a, q\right\},$$

then it is trivial that $E \subseteq F$. Now we will show $F \subseteq E$ to conclude $E = F$. Suppose by contradiction, there exists some x_0 such that $x_0 \in F$ s.t. $x_0 \notin E$, then by definition of F , there exist infinitely many positive integer pairs (a, q) s.t. $\left|x_0 - \frac{a}{q}\right| \leq \frac{c}{q^p}$. Also, since $x_0 \notin E$, there must exist infinitely many positive integer pairs (a, q) with $a > q$ s.t. $\left|x_0 - \frac{a}{q}\right| \leq \frac{c}{q^p}$. However, for each such (a, q) pair with $a > q$ s.t. $\left|x_0 - \frac{a}{q}\right| \leq \frac{c}{q^p}$, we have

$$\frac{c}{q^p} \geq \left|x_0 - \frac{a}{q}\right| = \frac{a}{q} - x_0 \geq \frac{a}{q} - 1 \geq \frac{1}{q}$$

since $x_0 \in [0, 1]$ and $a \geq q + 1$, and thus $c \geq q^{p-1}$. Now we claim that if $x_0 \in F$, then $\left|x_0 - \frac{a}{q}\right| \leq \frac{c}{q^p}$ for infinitely many positive integers q . If not, then for some q_0 , there exists infinitely many positive integers a s.t. $\left|x_0 - \frac{a}{q_0}\right| \leq \frac{c}{q_0^p}$. However, this is impossible since $\frac{c}{q_0^p}$ is a fixed value and

$$\left|x_0 - \frac{a}{q_0}\right| \leq \frac{c}{q_0^p} \Rightarrow x_0 \geq \frac{a}{q_0} - \frac{c}{q_0^p},$$

so if there are infinitely many positive integers a s.t. $|x_0 - \frac{a}{q_0}| \leq \frac{c}{q_0^p}$, then x_0 must be greater than or equal to infinitely many positive numbers, which is impossible since $x_0 \in [0, 1]$. Now we have proved our claim, and thus $c \geq q^{p-1}$ for infinitely many positive integers q , which is impossible since $p - 1 > 1$ and thus q^{p-1} will diverge when q gets large. Hence, we have $F \subseteq E$ and thus $E = F$. Therefore, we conclude that the set F has $m(F) = m(E) = 0$, and we're done. ■

Problem 2. Call a real number $x \in \mathbb{R}$ *diophantine* if there exist real numbers $p, C > 0$ such that $|x - \frac{a}{q}| > C/|q|^p$ for all nonzero integers q and all integers a . Using Exercise 8.2.6, show that almost every real number is diophantine. (Hint: first work in the interval $[0, 1]$. Show that one can take p and C to be rational and one can also take $p > 2$. Then use the fact that the countable union of measure zero sets has measure zero.)

Problem 3. For every positive integer n , let $f_n : \mathbb{R} \rightarrow [0, \infty)$ be a non-negative measurable function such that

$$\int_{\mathbb{R}} f_n \leq \frac{1}{4^n}.$$

Show that for every $\varepsilon > 0$, there exists a set E of Lebesgue measure $m(E) \leq \varepsilon$ such that $f_n(x)$ converges pointwise to zero for all $x \in \mathbb{R} \setminus E$. (Hint: first prove that $m(\{x \in \mathbb{R} : f_n(x) > \frac{1}{\varepsilon 2^n}\}) \leq \frac{\varepsilon}{2^n}$ for all $n = 1, 2, 3, \dots$, and then consider the union of all the sets $\{x \in \mathbb{R} : f_n(x) > \frac{1}{\varepsilon 2^n}\}$.)

Proof. Let $E_n := \{x \in \mathbb{R} : f_n(x) > \frac{1}{\varepsilon 2^n}\}$, then we know

$$\frac{1}{\varepsilon 2^n} \chi_{E_n} \leq f_n \quad \forall x \in \mathbb{R}.$$

Note that since for each positive integer n , f_n is measurable, so $E_n = f_n^{-1}(\frac{1}{\varepsilon 2^n}, \infty)$ is measurable and thus χ_{E_n} is measurable, which gives $\frac{1}{\varepsilon 2^n} \chi_{E_n}$ is measurable, and it is also non-negative. Thus, for each $n \in \mathbb{N}$, we have

$$\frac{1}{\varepsilon 2^n} m(E_n) = \int_{\mathbb{R}} \frac{1}{\varepsilon 2^n} \chi_{E_n} \leq \int_{\mathbb{R}} f_n \leq \frac{1}{4^n} \Rightarrow m(E_n) \leq \frac{\varepsilon}{2^n}.$$

Now define $E := \bigcup_{n=1}^{\infty} E_n$, then

$$\mathbb{R} \setminus E = \left\{ x \in \mathbb{R} : f_n(x) \leq \frac{1}{\varepsilon 2^n} \quad \forall n \in \mathbb{N} \right\}.$$

Hence, $\lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in \mathbb{R} \setminus E$. Also,

$$m(E) = m\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \varepsilon,$$

so we're done. ■

Problem 4. For every positive integer n , let $f_n : [0, 1] \rightarrow [0, \infty)$ be a non-negative measurable function such that f_n converges pointwise to zero. Show that for every $\varepsilon > 0$, there exists a set E of Lebesgue measure $m(E) \leq \varepsilon$ such that $f_n(x)$ converges *uniformly* to zero for all $x \in [0, 1] \setminus E$. (This is a special case of Egoroff's theorem. To prove it, first show that for any positive integer m , we can find an $N > 0$ such that $m(\{x \in [0, 1] : f_n(x) > 1/m \text{ for all } n \geq N\}) \leq \varepsilon/2^m$.) Is the claim still true if $[0, 1]$ is replaced by $\mathbb{R}$?

Problem 5. Give an example of a bounded non-negative function $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}^+$ such that $\sum_{m=1}^{\infty} f(n, m)$ converges for every n , and such that $\lim_{n \rightarrow \infty} f(n, m)$ exists for every m , but such

that

$$\lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} f(n, m) \neq \sum_{m=1}^{\infty} \lim_{n \rightarrow \infty} f(n, m).$$

(Hint: modify the moving bump example. It is even possible to use a function f which only takes the values 0 and 1. This shows that interchanging limits and infinite sums can be dangerous.)

Proof. Define

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}^+, \quad f(n, m) = \begin{cases} 1, & \text{if } m = n; \\ 0, & \text{if } m \neq n. \end{cases}$$

then for each n , $\sum_{m=1}^{\infty} f(n, m) = 1$, and for each m , $\lim_{n \rightarrow \infty} f(n, m) = 0$. Hence,

$$\lim_{n \rightarrow \infty} \sum_{m=1}^{\infty} f(n, m) = 1 \text{ and } \sum_{m=1}^{\infty} \lim_{n \rightarrow \infty} f(n, m) = 0,$$

which is not equal, so we're done. ■

Problem 6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be absolutely integrable, measurable functions such that $f(x) \leq g(x)$ for all $x \in \mathbb{R}$, and that $\int_{\mathbb{R}} f = \int_{\mathbb{R}} g$. Show that $f(x) = g(x)$ for almost every $x \in \mathbb{R}$ (i.e., that $f(x) = g(x)$ for all $x \in \mathbb{R}$ except possibly for a set of measure zero).

Problem 7. For each $n = 1, 2, 3, \dots$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be the function $f_n = \chi_{[n, n+1)} - \chi_{[n+1, n+2)}$; i.e., let $f_n(x)$ equal +1 when $x \in [n, n+1)$, equal -1 when $x \in [n+1, n+2)$, and 0 everywhere else. Show that

$$\int_{\mathbb{R}} \sum_{n=1}^{\infty} f_n \neq \sum_{n=1}^{\infty} \int_{\mathbb{R}} f_n.$$

Explain why this does not contradict Corollary 8.2.11.

Problem 8 (Tannery's convergence theorem for Riemann integrals). Prove Tannery's convergence theorem for Riemann integrals.

Let $\{f_n\}$ be a sequence of functions and let $\{p_n\}$ be an increasing sequence of real numbers such that

$$p_n \rightarrow +\infty \quad \text{as } n \rightarrow \infty.$$

Assume that:

- (a) $f_n \rightarrow f$ uniformly on $[a, b]$ for every $b \geq a$.
- (b) f_n is Riemann integrable on $[a, b]$ for every $b \geq a$.
- (c)

$$|f_n(x)| \leq g(x)$$

almost everywhere on $[a, +\infty)$, where g is nonnegative and improper Riemann integrable on $[a, +\infty)$.

Then both f and $|f|$ are improper Riemann integrable on $[a, +\infty)$, the sequence

$$\left\{ \int_a^{p_n} f_n(x) dx \right\}$$

converges, and

$$\int_a^{+\infty} f(x) dx = \lim_{n \rightarrow \infty} \int_a^{p_n} f_n(x) dx.$$

(d) Use Tannery's theorem to prove that, if $p > -1$, then

$$\lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{x}{n}\right)^n x^p dx = \int_0^\infty e^{-x} x^p dx.$$

Hint: Use the relation between Riemann and Lebesgue integrals. Namely, recall that if a function is Riemann integrable on a bounded interval, then it is Lebesgue integrable there, and the two integrals agree.

For the main theorem, define

$$F_n(x) := f_n(x)\chi_{[a, p_n]}(x), \quad x \in [a, \infty).$$

Since $p_n \rightarrow \infty$, for each fixed $x \geq a$, we eventually have $x \leq p_n$. Hence

$$F_n(x) \rightarrow f(x) \quad \text{a.e. on } [a, \infty).$$

Moreover,

$$|F_n(x)| = |f_n(x)|\chi_{[a, p_n]}(x) \leq g(x) \quad \text{a.e. on } [a, \infty).$$

Since g is nonnegative and improper Riemann integrable on $[a, \infty)$, it is Lebesgue integrable on $[a, \infty)$. Therefore the Lebesgue dominated convergence theorem gives

$$\lim_{n \rightarrow \infty} \int_a^\infty F_n(x) dx = \int_a^\infty f(x) dx.$$

Finally, observe that

$$\int_a^\infty F_n(x) dx = \int_a^{p_n} f_n(x) dx.$$

For part (d), apply the theorem with

$$p_n = n, \quad f_n(x) = \left(1 - \frac{x}{n}\right)^n x^p \quad \text{on } [0, n].$$

Use the pointwise limit

$$\left(1 - \frac{x}{n}\right)^n \rightarrow e^{-x}$$

and find an integrable majorant for

$$\left|\left(1 - \frac{x}{n}\right)^n x^p\right|.$$

A useful estimate is

$$0 \leq \left(1 - \frac{x}{n}\right)^n \leq e^{-x}, \quad 0 \leq x \leq n.$$

Since

$$e^{-x} x^p \in L^1([0, \infty)) \quad \text{for } p > -1,$$

the dominated convergence theorem applies.